Assignment 3

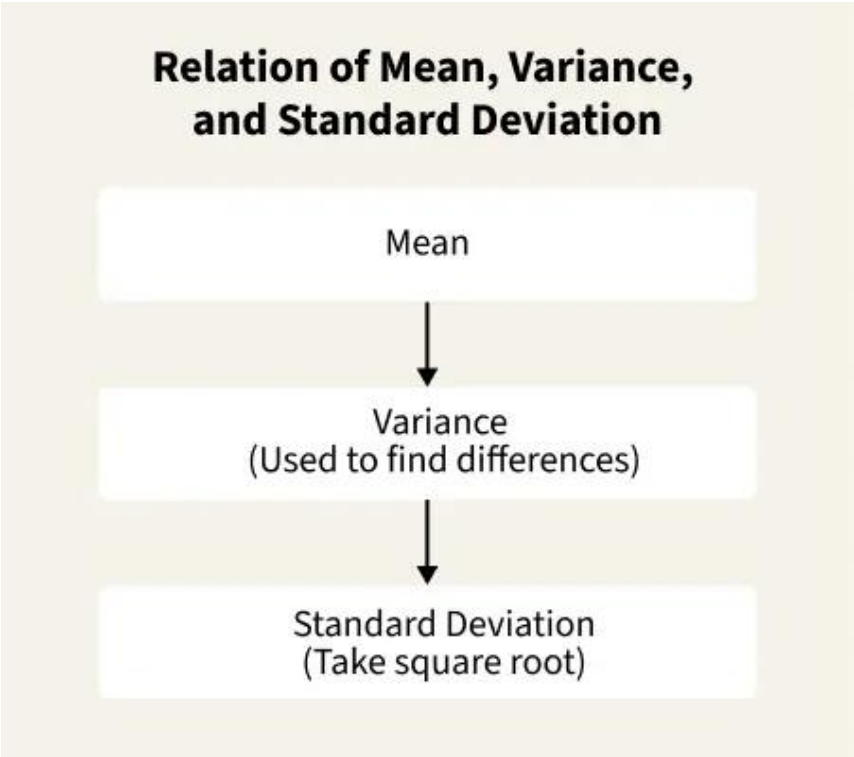
"If μ= 55, σ_{4a}= 4, $\sigma_{4\beta}$=10, σ_4=c 15, In this which is better?"

Concept of Mean and Standard Deviation:

In statistics, when we analyze any data such as students' exam scores, we need some basic measures to understand the data properly. Two very important statistical terms used for this purpose are Mean and Standard Deviation. These measures help us understand the average performance and how much the data varies from that average.

First, let us understand the Mean.:

Mean is also called the average. It tells us the central value of a data set. It is calculated by adding all the values and dividing the total by the number of values.



Formula for Mean:

$$\text{Mean } (\mu) = \frac{\text{Sum of all values}}{\text{Number of values}}$$

For example, suppose five students scored 50, 60, 55, 45 and 65 marks.

Step 1: Add all marks
50 + 60 + 55 + 45 + 65 = 275

Step 2: Divide by total number of students (5)

Mean = 275 / 5 = 55

So, the average score is 55. This means the central performance of the class is 55 marks.

Importance of Mean:
The mean helps us to understand the overall level of performance. It gives one single value that represents the whole data. In exams, it tells whether the class performance is low, average, or high.

However, mean alone is not enough. It does not tell us how the marks are spread around the average. For that, we use Standard Deviation.

Now let us understand Standard Deviation.

Standard Deviation (σ) measures how much the data values differ from the mean. It tells us about the spread or variability of data.

If the standard deviation is small, it means the data values are close to the mean.
If the standard deviation is large, it means the data values are widely spread from the mean.

In simple words:
Small σ → More consistent performance
Large σ → Less consistent performance

Formula for Standard Deviation:

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{N}}$$

Where:
xi= each value
μ = mean
N = total number of values

Let us take a simple example.

Suppose the mean score of a class is 55.

Case 1: Scores are 53, 54, 55, 56, 57
These scores are very close to 55. So standard deviation will be small.

Case 2: Scores are 30, 40, 55, 70, 80
These scores are far from 55. So standard deviation will be large.

Even though both cases may have same mean (55), their spread is different. This difference is measured by standard deviation.

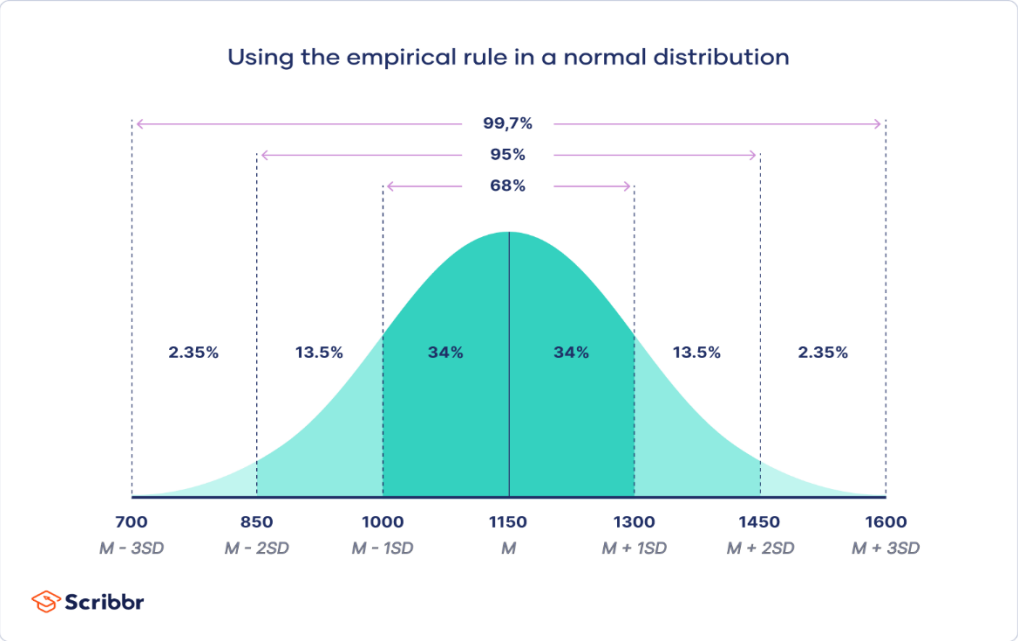
Importance of Standard Deviation:

Standard deviation helps to:
• Understand consistency of performance
• Compare different groups

• Identify variation in data
• Detect outliers (very high or very low values)

Understanding Normal Distribution:

Normal distribution is a continuous probability distribution that shows how data values are distributed around the mean. It forms a symmetric, bell-shaped curve. In this distribution, most of the values are concentrated near the mean, and fewer values are found at the extremes.



- **Bell-Shaped Curve**
  When data follows normal distribution, its graph forms a smooth bell-shaped curve. The curve rises gradually, reaches a peak at the mean, and then decreases symmetrically on both sides.

- **Symmetry About the Mean**
  The curve is perfectly symmetric. This means the left side and right side of the curve are mirror images. Half of the data lies on the left of the mean and half lies on the right.

- **Mean = Median = Mode**
  In a perfectly normal distribution, the mean, median, and mode are equal. All three measures of central tendency lie at the center of the distribution.

- **Concentration of Values**
  Most of the data values lie close to the mean. Only a small number of values are very high or very low. This shows that extreme values are rare.

- **Total Area Under the Curve**
  The total area under the normal curve is equal to 1 or 100%. This represents the total probability of all possible values.

- **Effect of Standard Deviation**
  The shape of the curve depends on standard deviation. A small standard deviation makes the curve narrow and tall. A large standard deviation makes it wide and flat.

- **Application in Exam Scores**
  Students' exam scores often follow normal distribution. Most students score around the average, while very few score extremely high or extremely low.

Given Data and Problem Statement($\mu = 55$, $\sigma A = 4$, $\sigma B = 10$, $\sigma C = 15$):
The given problem provides three normal distributions with the same mean but different standard deviations. The objective is to compare these distributions and determine which one shows better and more consistent performance.

- **Same Mean for All Cases**
  The mean (μ) is 55 in all three cases. This means the average score of students is 55 in each group.

- **Different Standard Deviations**
  The standard deviation values are different:
  $\sigma A = 4$
  $\sigma B = 10$
  $\sigma C = 15$
  This means the spread of marks is different in each case.

- **Center of the Curve**
  Since the mean is 55 for all three, the center of all three normal curves will be at 55.

- **Difference in Spread**
  Although the center is the same, the width of the curves will differ because of different standard deviations.

- **Objective of the Problem**
  The main aim is to compare the three distributions and decide which one is better based on consistency and variation.

- **Basis of Comparison**
  Because the mean is same, we cannot compare using average performance. Therefore, we must compare using standard deviation.

Role of Standard Deviation in Comparing Performance
Standard deviation is a statistical measure that shows how much the data values vary from the mean. It plays an important role in comparing consistency when the average values are the same.

- **Measure of Variation**
  Standard deviation measures how far individual values are from the mean. It tells us whether the data is tightly clustered or widely spread.

- **Small Standard Deviation**
  When standard deviation is small, most values are very close to the mean. This indicates stable and consistent performance.

- **Large Standard Deviation**
  When standard deviation is large, values are spread far from the mean. This indicates high variation and less consistency.

- **Comparison When Mean is Same**
  If two or more groups have the same mean, standard deviation becomes the deciding factor for comparison.

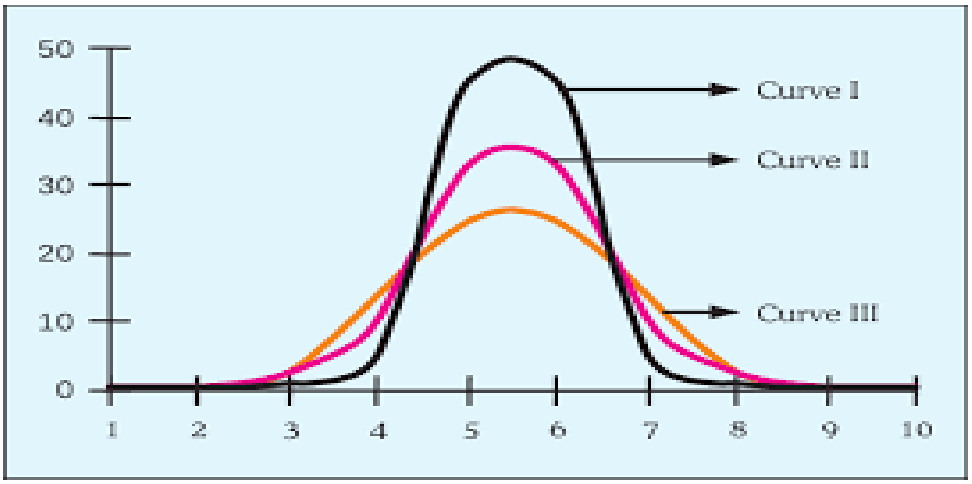
- **Case A ($\sigma = 4$)**
  This shows very small variation. Students' scores are very close to 55. Performance is highly consistent.

- **Case B ($\sigma = 10$)**
  This shows moderate variation. Scores are somewhat spread out. Performance consistency is average.

- **Case C ($\sigma = 15$)**
  This shows large variation. Scores are widely spread. Some students may score very high and some very low.

- **Decision**
  Among the three, Distribution **A ($\sigma = 4$) is better** because it **shows least variation and highest consistency in performance.**

Graphical Representation – Comparison of Three Normal Curves:



Explanation of the Graph:

The given graph shows three normal distribution curves labeled as Curve I, Curve II, and Curve III. All three curves are centered at the same point on the horizontal axis, which represents the Mean (μ). This means the average value for all three distributions is the same.

The horizontal axis represents the data values (such as exam scores), and the vertical axis represents the frequency or probability.

Now let us understand each curve:

Curve I (Tall and Narrow Curve):
This curve is the tallest and narrowest among the three. It has the smallest spread. This means the standard deviation is small. When standard deviation is small, most of the values are very close to the mean. So, the performance is very consistent. In your given data, this corresponds to $\sigma = 4$.

Curve II (Medium Spread Curve);
This curve is moderately tall and moderately wide. It has a medium spread. This means the standard deviation is moderate. The data values are somewhat spread around the mean. Performance consistency is average. This corresponds to $\sigma = 10$**.**

Curve III (Wide and Flat Curve):
This curve is the widest and flattest. It has the largest spread. This means the standard deviation is large. The values are widely spread from the mean. Some values are very high and some are very low. Performance consistency is low. This corresponds to $\sigma = 15$.

Important Observations from the Graph:

- All three curves have the same mean because they peak at the same center point.
- The difference between the curves is due to different standard deviations.
- Smaller standard deviation makes the curve taller and narrower.
- Larger standard deviation makes the curve shorter and wider.
- The area under all three curves is equal (100%).

Analysis of Variability and Consistency:
Variability shows how much the data values spread from the mean. Consistency shows how closely the values are grouped around the mean. Standard deviation measures this spread.

- All three distributions have mean 55.
- $\sigma = 4$ shows very small variation.
- $\sigma = 10$ shows moderate variation.
- $\sigma = 15$ shows high variation.
- Smaller standard deviation means higher consistency.
- Larger standard deviation means lower consistency.
- Distribution A is the most consistent.

Interpretation – Which Distribution is Better and Why?
When the mean is same, the distribution with smaller standard deviation is considered better because performance is more stable.

• Average score is equal in all cases.
• Comparison depends on standard deviation.
• Distribution A ($\sigma = 4$) has least spread.
• Distribution B ($\sigma = 10$) has moderate spread.
• Distribution C ($\sigma = 15$) has highest spread.
• Least spread means most balanced performance.
• Therefore, Distribution A is better.

Real-Life Application in Students' Exam Scores:
Mean and standard deviation are used in schools to analyze and compare student results.

• Mean shows average marks of the class.
• Standard deviation shows performance variation.
• Small σ means students performed similarly.
• Large σ means big difference between students.
• Helps teachers improve teaching methods.

Conclusion:
Standard deviation plays a key role when the mean is same.

• Mean = 55 for all distributions.
• $\sigma = 4$ has least variability.
• $\sigma = 10$ has moderate variability.
• $\sigma = 15$ has highest variability.
• Lower variability indicates better consistency.
• Hence, Distribution A ($\sigma = 4$) is the best.